

Dear Editors,

Please consider my manuscript, “Observation-conditioned bounds enable experimental decisions despite mechanistic equivalence in bacterial nitrogen signaling,” for publication as a Research Article in *PLOS Computational Biology*. The work addresses a central problem in computational biology: how mechanistic models can guide experiments when distinct molecular explanations reproduce the same biological response. Its central advance is an observation-conditioned guarantee that converts mechanistic non-identifiability into a quantitative concentration decision, together with an end-to-end benchmark showing when complementary binding measurements can and cannot resolve the ambiguity.

Using the *Escherichia coli* GlnD–PII–GlnE–glutamine synthetase cascade, the study makes three linked contributions. It constructs biochemical mechanism families that preserve complete stationary target-state distributions while predicting different enzyme occupancies and kinetics. It then combines protein conservation with a concurrent PII modification-count constraint to tighten the allowed total-GS difference from 0.370 to 0.041, crossing a prespecified 0.05 tolerance and certifying that a fivefold reduction in GlnE concentration keeps the model-class discrepancy below that tolerance. Finally, it follows a complementary binding experiment from prepared doses through ligand-depletion-aware fitting to statistical discrimination: independent concentration standards correct the tested systematic bias, while an unmodeled affinity mixture can still produce false mechanistic attribution despite a convincing fit.

The study moves beyond showing that several models fit: it identifies predictions shared across an equivalent model class, converts an added observation into an assay-level decision, and tests the measurement process needed to separate models within that class. This integration of structural non-identifiability, biochemical network modeling and quantitative experiment design should interest the broad systems-biology readership of *PLOS Computational Biology* and is transferable to other multisite modification cascades governed by bifunctional enzymes.

The study is deliberately model-class conditional and uses published measurements to define a prospective experimental workflow. The manuscript clearly separates class-wide guarantees from claims about native biology and specifies the matched-assay measurements required for prospective validation. Data, executable analyses, independent numerical checks and complete manuscript sources accompany the submission.

If available, I suggest Julio R. Banga (Computational Biology Lab, MBG-CSIC) as Academic Editor because of his expertise in mechanistic systems biology, model identifiability, uncertainty quantification and experimental design.

This manuscript is original, has not been published, and is not under consideration elsewhere. I am the sole author and declare no competing interests. All published data and software reused in the study are cited and their provenance and licenses are documented.

Thank you for considering the manuscript.

Sincerely,

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